

Information Retrieval and Web Search, Fall 2025

Assignment 1 - Spelling Error Correction, Due: 02 November 2025 23:59

In this assignment you will implement a simple spelling error corrector. Before you start this assignment, I suggest you to read the article by Peter Norvig on “How to Write a Spelling Corrector” available at <http://norvig.com/spell-correct.html>. The spelling error corrector that you will implement will be simpler than the one described in this article.

First, you should create a dictionary containing the English words and their frequencies by using the provided `corpus.txt` file. This file was obtained from Peter Norvig’s web site. It contains a concatenation of several public domain books from Project Gutenberg and lists of most frequent words from Wiktionary and the British National Corpus. You can assume that the words in the `corpus.txt` file are spelled correctly. In order to create your dictionary, you will need to tokenize the file and perform case-folding. You can predict the correct spelling of a misspelled word by generating all the words whose edit distances to the word are 1 and select the one with the highest frequency in the `corpus.txt` file. Note that several spelling errors involve transpositions of characters. Therefore, you should use the Damerau-Levenshtein edit distance, where the valid operations are defined as insertion, deletion, and substitution of a single character, or transposition of two adjacent characters.

You can use any programming language. The program should take a file containing a list of misspelled words (one word per line) as input, and produce a file with the predicted correct spellings of these words (one word per line) as output. If your program cannot produce predictions for any of the words in the input file, the corresponding lines in the output file should be printed as blank lines.

A list of 384 misspelled words (`test-words-misspelled.txt`) and their corresponding correct spellings (`test-words-correct.txt`) are provided for you to test your program. Compute the accuracy of your program for this test set and write it in your report. **Suggest and implement an extension for your spelling corrector.** Provide the accuracy results obtained by the enhanced algorithm in your report. The extension that you suggest can be a simple heuristic such as assuming that the first letter of a word is always spelled correctly.

Submission: You should submit a “.zip” file named as `YourStudentID.zip` containing the following files through LMS:

1. *Report*: Describe the extension that you propose for the baseline spelling corrector. Report the **accuracy** results obtained by both versions of your spelling corrector on the provided test set.
2. *Source code and executable*: Commented source code and executables of both versions of your spelling corrector.
3. *Readme*: Describing how to run your program. We should be able to run your program using a different test set.

CAUTION: You should write your own code for the edit distance algorithm according to the pseudocodes that have already been shared with you in the lecture slides. **THE USAGE OF ANY EXTERNAL LIBRARY IS NOT ALLOWED.**